

SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE
SUBJECT - Data Analysis with R/SAS/SPSS

Aim - Performing paired t-tests using `t.test(paired=TRUE)` (R).

Output :

The image displays two screenshots of the RStudio interface, showing the execution of R code and the resulting output in the console.

Top Screenshot:

```
Source
Console Terminal Background Jobs
R - R4.52 - / -
$ installment : num 581.9 573.2 76.3 468.1 395.5 ...
$ grade_subgrade : chr "B5" "F1" "B4" "A5" ...
$ num_of_open_accounts : int 7 5 2 7 1 3 6 4 6 4 ...
$ total_credit_limit : num 40833 27968 15502 18158 17468 ...
$ current_balance : num 24302 10803 4505 5526 3594 ...
$ delinquency_history : int 1 1 0 4 2 0 3 0 3 2 ...
$ public_records : int 0 0 0 0 0 0 0 0 0 0 ...
$ num_of_delinquencies : int 1 3 0 5 2 1 3 0 4 2 ...
$ loan_paid_back : int 1 1 1 1 1 1 0 1 1 0 ...
> summary(data$annual_income)
Min. 1st Qu. Median Mean 3rd Qu. Max.
6000 24261 36585 43550 54678 400000
> summary(data$monthly_income)
Min. 1st Qu. Median Mean 3rd Qu. Max.
500 2022 3049 3629 4556 33333
> summary(data$loan_amount)
Min. 1st Qu. Median Mean 3rd Qu. Max.
500 8853 14946 15129 20999 49040
> summary(data$installment)
Min. 1st Qu. Median Mean 3rd Qu. Max.
9.43 253.91 435.60 455.63 633.60 1685.40
> t.test(data$annual_income, data$monthly_income, paired = TRUE)

Paired t-test

data: data$annual_income and data$monthly_income
t = 214.83, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 39556.27 40284.73
sample estimates:
mean difference
 39920.5
```

Bottom Screenshot:

```
Source
Console Terminal Background Jobs
R - R4.52 - / -
t.test(data$annual_income, data$monthly_income, paired = TRUE, alternative = "greater")

Paired t-test

data: data$annual_income and data$monthly_income
t = 214.83, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 39614.83      Inf
sample estimates:
mean difference
 39920.5

> t.test(data$annual_income, data$monthly_income, paired = TRUE, alternative = "less")

Paired t-test

data: data$annual_income and data$monthly_income
t = 214.83, df = 19999, p-value = 1
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
      -Inf 40226.17
sample estimates:
mean difference
 39920.5

> t.test(data$annual_income, data$monthly_income, paired = TRUE, conf.level = 0.95)

Paired t-test

data: data$annual_income and data$monthly_income
t = 214.83, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 39556.27 40284.73
sample estimates:
mean difference
 39920.5
```

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SUBJECT - Data Analysis with R/SAS/SPSS

The screenshot displays the RStudio interface with the following components:

- Source Panel:** Contains R code for three t-tests:

```
> t.test(data$loan_amount, data$installment, paired = TRUE)

Paired t-test

data: data$loan_amount and data$installment
t = 248.63, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 14557.99 14789.36
sample estimates:
mean difference
 14673.68

> t.test(data$loan_amount, data$installment, paired = TRUE, alternative = "greater")

Paired t-test

data: data$loan_amount and data$installment
t = 248.63, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 14576.59      Inf
sample estimates:
mean difference
 14673.68

> t.test(data$loan_amount, data$installment, paired = TRUE, alternative = "less")

Paired t-test

data: data$loan_amount and data$installment
t = 248.63, df = 19999, p-value = 1
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
      -Inf 14770.76
sample estimates:
mean difference
 14673.68
```
- Console Panel:** Shows the output of the first t-test, which is identical to the code block above.
- Environment Panel:** Displays the 'data' object with 20000 observations and 22 variables.
- Files Panel:** Shows a file explorer view of the user's home directory, listing various files and folders.

The second screenshot shows the RStudio interface with the following components:

- Source Panel:** Contains R code for three t-tests:

```
> t.test(data$loan_amount, data$installment, paired = TRUE, conf.level = 0.95)

Paired t-test

data: data$loan_amount and data$installment
t = 248.63, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 14557.99 14789.36
sample estimates:
mean difference
 14673.68

> t.test(data$annual_income, data$loan_amount, paired = TRUE)

Paired t-test

data: data$annual_income and data$loan_amount
t = 134.47, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 28006.06 28834.61
sample estimates:
mean difference
 28420.34

> t.test(data$monthly_income, data$installment, paired = TRUE)

Paired t-test

data: data$monthly_income and data$installment
t = 186.71, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 3140.195 3206.826
sample estimates:
mean difference
 3173.511
```
- Console Panel:** Shows the output of the first t-test, which is identical to the code block above.
- Environment Panel:** Displays the 'data' object with 20000 observations and 22 variables.
- Files Panel:** Shows a file explorer view of the user's home directory, listing various files and folders.

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